$$v = \frac{(60)(0.81)}{40} = 23.54 = 24 \text{ m s}^{-1} \text{ (2 s.f)}$$

6 **C** Impulse = change in momentum = final momentum – initial momentum
Taking the direction away from the player as the positive direction,

$$+6.7 = 0.45v_f - 0.45(-3.2)$$

$$v_f = 12 \text{ m s}^{-1}$$